

The Illusion of Inference

A Neuro-Symbolic Falsification of LLM Reasoning via Formal Verification and Token Surprisal

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The Core Problem: True Reasoning vs. Pattern Matching

- **The LLM Promise:** Modern autoregressive models (Qwen, Gemma, Mistral) appear to solve complex natural language logic problems.
- **The Underlying Question:** Are these models executing *formal deductive rules*, or are they merely *stochastic parrots* retrieving statistical co-occurrences?
- **Formal Invariance Principle:** In First-Order Logic (FOL), an inference $M \models \alpha$ is valid strictly based on syntax, completely invariant to entity symbols.

Central Research Goal

Build a **falsifiable Neuro-Symbolic framework** that decouples structural logic from semantic priors to prove LLMs mimic reasoning rather than execute it.

Falsifiable Hypotheses

H1: Invariance to Surface Perturbations

LLM logical reasoning performance drops significantly when evaluating logically equivalent inferences featuring altered, nonsensical entity tokens (removing prior knowledge).

H2: Systematic Quantifier Scoping

LLMs systematically fail to distinguish between logical formulations with swapped quantifiers ($\forall x \exists y$ vs. $\exists y \forall x$) possessing nearly identical surface structures.

4-Step Methodology

- 1 **Data Neutralization (Caroline Filter):** Replace real entities (PERSON, ORG, GPE) in the FOLIO dataset with synthetic Lewis Carroll tokens via spaCy NER.
- 2 **Multi-Model LLM Inference:** Query 3 distinct LLM families (Qwen2.5, Gemma-2, Mistral) under zero-shot, greedy decoding (`do_sample=False`).
- 3 **Formal Logic Verification (Z3 Prover):** Apply Proof by Contradiction in Z3 SMT solver to mathematically certify logical invariance.
- 4 **Token Surprisal Extraction:** Extract raw PyTorch logits ($-\log P$) on 50 unique synthetic minimal pairs testing quantifier scope ambiguity.

Step 1: Semantic Neutralization (The Caroline Filter)

- **Objective:** Remove parametric memory biases without modifying logical syntax.
- **Implementation:** Extracted entities from FOLIO (*Han et al., 2022*) and applied a bijective mapping $f: E \rightarrow S$ to synthetic tokens.

Semantic Neutralization Example

Original FOLIO Premise:

"Tom is a wild turkey... Tom is not an Eastern wild turkey."

Caroline Filtered Premise:

"Boroogove is a wild turkey... Boroogove is not a Rath wild turkey."

Step 3: Formal Verification Loop via Z3 SMT Solver

- **Why Z3?** We must prove the Caroline filter did NOT corrupt the formal logic.
- **Proof by Contradiction:** We assert premises Φ and the **negation of conclusion** α to Z3:

$\text{CheckSatisfiability}(\Phi \wedge \neg\alpha)$

- **Result:** Z3 returned UNSAT for both Original and Caroline FOL formulations (*De Moura & Bjørner, 2008*).

Mathematical Guarantee

Z3 mathematically certifies that $\Phi_{orig} \models \alpha_{orig}$ and $\Phi_{caroline} \models \alpha_{caroline}$ are **100% logically identical**. Any drop in LLM accuracy is purely an architectural failure.

Step 4: Probing Quantifier Scoping via Token Surprisal

- **Scope Ambiguity:** Distributive ($\forall x \exists y$, e.g., multiple trees) vs. Cumulative ($\exists y \forall x$, e.g., single tree).
- **Minimal Pairs Design:** 50 strictly unique pairs generated using 10 templates and 5 adjective modifiers (*massive, huge, gigantic, large, enormous*).
- **Raw PyTorch Logit Extraction:** Bypassed discrete generation text to avoid surface noise.

$$\mathcal{S}(T \mid C) = - \sum_{i=1}^k \log P(t_i \mid C, t_1, \dots, t_{i-1}) = -\text{sum}(\log_softmax(\text{Shifted Logits}))$$

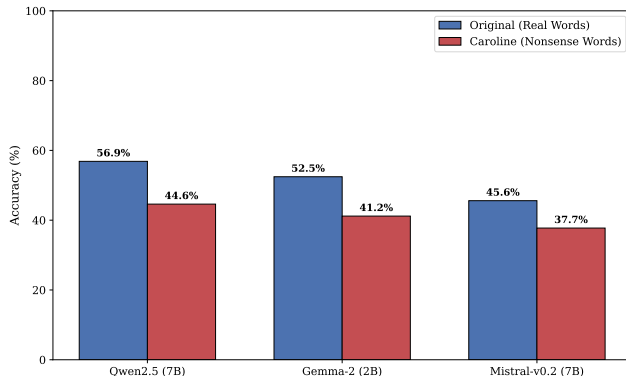
Experimental Setup & Reproducibility

Model Family	Developer	Parameters	Decoding Setting
Qwen2.5-7B-Instruct	Alibaba	7 Billion	Greedy (do_sample=False)
Gemma-2-2B-It	Google	2 Billion	Greedy (do_sample=False)
Mistral-7B-Instruct-v0.2	Mistral AI	7 Billion	Greedy (do_sample=False)

- **Hardware:** Kaggle Environment, Dual NVIDIA T4 GPUs (30GB total VRAM).
- **Precision:** Loaded in bfloat16 with `device_map="auto"`.
- **Memory Management:** Explicit `gc.collect()` and `torch.cuda.empty_cache()` invoked between models to guarantee non-interfering, reproducible runs.

H1 Results: Accuracy Drop Across LLM Families

H1: Logical Reasoning Accuracy Drop Across LLM Families
(Verified Logically Identical by Z3 Solver)



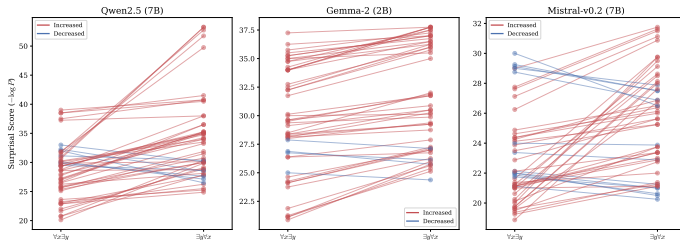
Empirical Findings (H1: Proven)

- **Qwen2.5 (7B):** Accuracy fell from 56.9% to 44.6% ($\Delta = -12.3\%$).
- **Gemma-2 (2B):** Accuracy fell from 52.5% to 41.2% ($\Delta = -11.3\%$).
- **Mistral-v0.2 (7B):** Accuracy fell from 45.6% to 37.8% ($\Delta = -7.8\%$).

Key Takeaway: All models collapse toward chance performance when stripped of entity familiarity.

H2 Results: Surprisal Inconsistency in Quantifier Scoping

H2: Surprisal Inconsistency in Quantifier Scoping Across LLM Families
(Lack of Systematic Logical Differentiation)



Empirical Findings (H2: Proven)

Swapping quantifiers ($\forall x \exists y \leftrightarrow \exists y \forall x$) yields a **chaotic, non-systematic spaghetti plot** across all 3 model families. Models react to lexical token co-occurrence (e.g., dog/bone vs. cat/mouse) rather than formal quantifier syntax.

Discussion & Methodological Limitations

- **Semantic Parsing Bottleneck:** Unrestricted NL-to-FOL translation remains an open problem. Automatically running Z3 across all 204 prompts introduces LLM translation errors.
- **Our Solution:** A *Neuro-Symbolic Case Study* pairing formal manual Z3 modeling with complete empirical LLM evaluation.
- **Negative Result (Architectural Variance):** Models exhibit varying failure magnitudes—Qwen drops 12.25%, Gemma drops 11.27%, and Mistral drops 7.84%.
- **Tokenization Artifacts:** Sub-word BPE tokenization fragments perturbed phrasing, conflating syntactic attention shifts with logical failures.

Main Conclusion

Autoregressive LLMs do **not** perform formal deductive reasoning. They act as sophisticated stochastic pattern matchers whose apparent reasoning collapses under semantic perturbation and quantifier swapping.

Future Paradigm Shift

Pure neural scaling cannot solve fundamental logical deficiencies. Future artificial general intelligence must pivot toward **hybrid Neuro-Symbolic architectures** pairing LLM natural language understanding with deterministic SMT/SAT formal provers.

Thank you for your attention!

All Source Code, Datasets, and Vector Artifacts are Open-Source:

<https://github.com/alireza-ebrahimi-ai/LLM-Logical-Reasoning-Evaluation>

Ready for the Oral Defense and Questions.